

Delivery Delay Prediction

1. Executive Summary

This report summarizes the data preparation, exploratory analysis, delivery-delay classification, and customer-rating prediction implemented in the submitted notebook. The analytical master dataset contains 96,478 orders, of which 91.89% were on time and 8.11% were late. The project uses delivery delay as the primary business target and also extends the analysis to predicting customer review ratings.

The classification results show a clear trade-off caused by class imbalance. The balanced Logistic Regression model achieved the strongest recall among the tested delay classifiers (61.73%) and the best F1 score (0.228) among the listed models. Random Forest produced the highest ROC-AUC (0.743), but its recall for late orders was only 1.41%. Therefore, the balanced Logistic Regression is the more practical model when the business priority is identifying as many potentially late orders as possible.

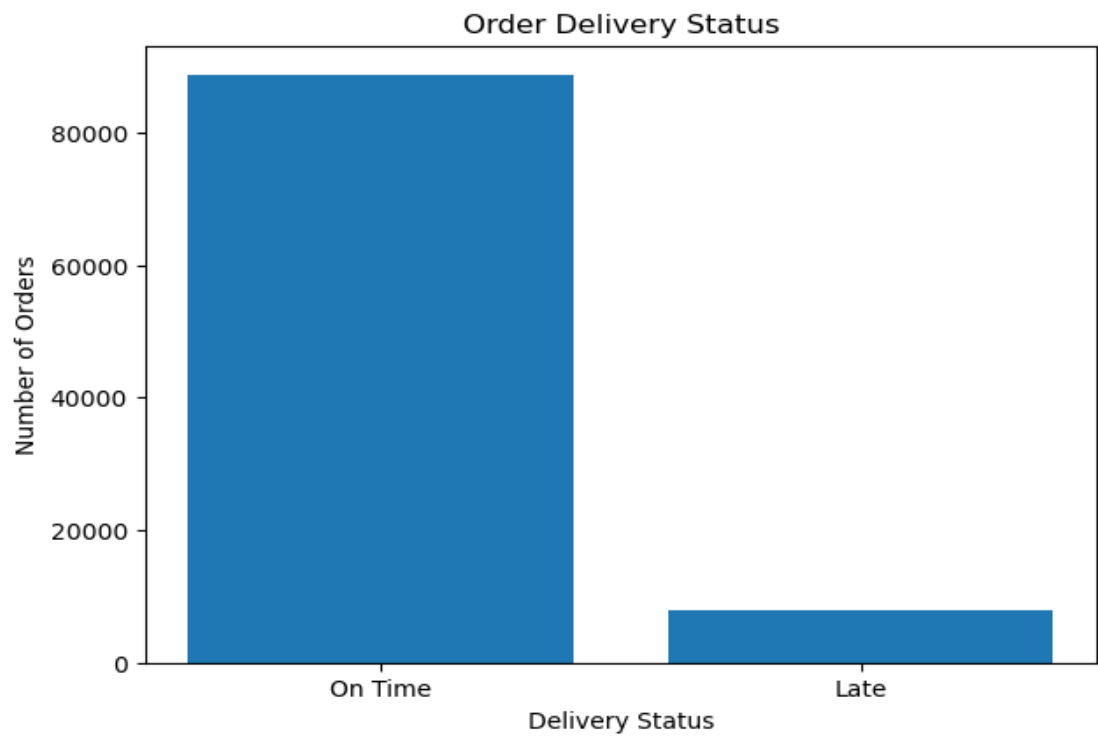
2. Data Sources & Data Quality

The notebook integrates nine Olist datasets: orders, customers, order items, payments, reviews, products, sellers, geolocation, and product-category translation. It performs schema inspection, missing-value checks, duplicate checks, key audits, referential-integrity checks, and relationship-cardinality analysis before constructing the order-level master table.

Metric	Notebook result
Master dataset	96,478 rows × 36 columns
On-time orders	88,652 (91.89%)
Late orders	7,826 (8.11%)
Delay-model train/test	77,182 / 19,296
Rating dataset	95,832 rows
Rating-model train/test	76,665 / 19,167

3. Exploratory Analysis

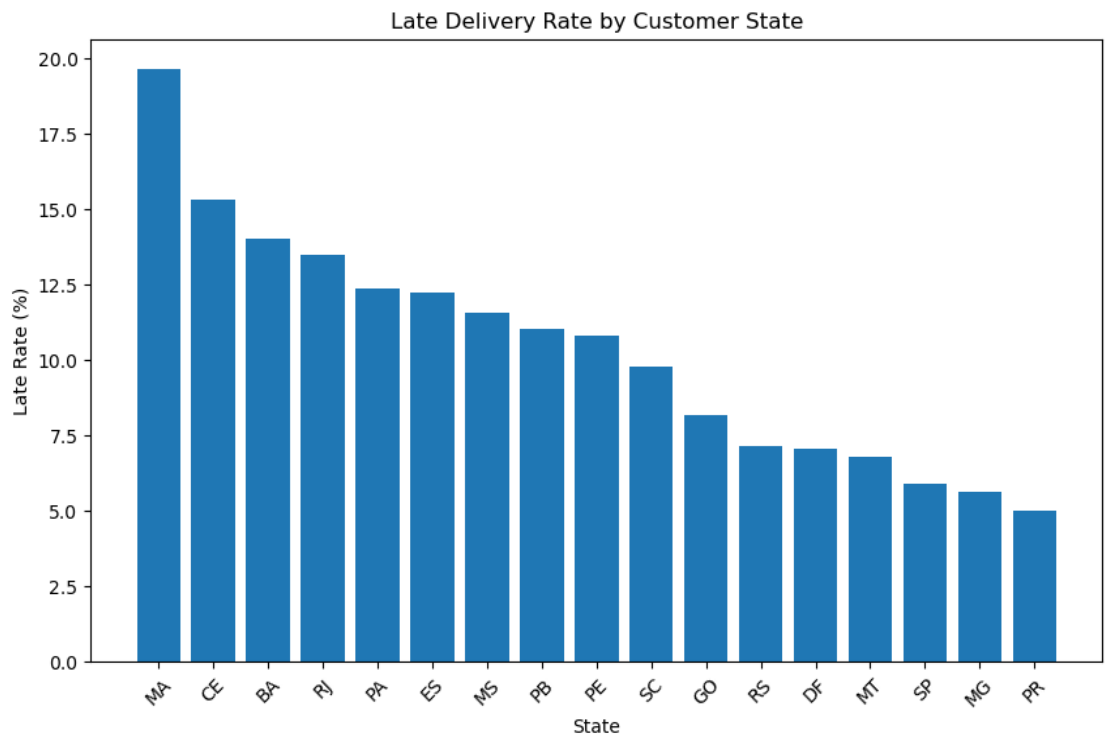
Late orders have higher average monetary and processing values than on-time orders. Late orders averaged 148.09 in item price versus 136.07 for on-time orders, 24.62 in freight versus 22.62, and 12.31 approval-delay hours versus 10.10. Conversely, the estimated delivery window was shorter for late orders (21.97 days versus 23.89 days).



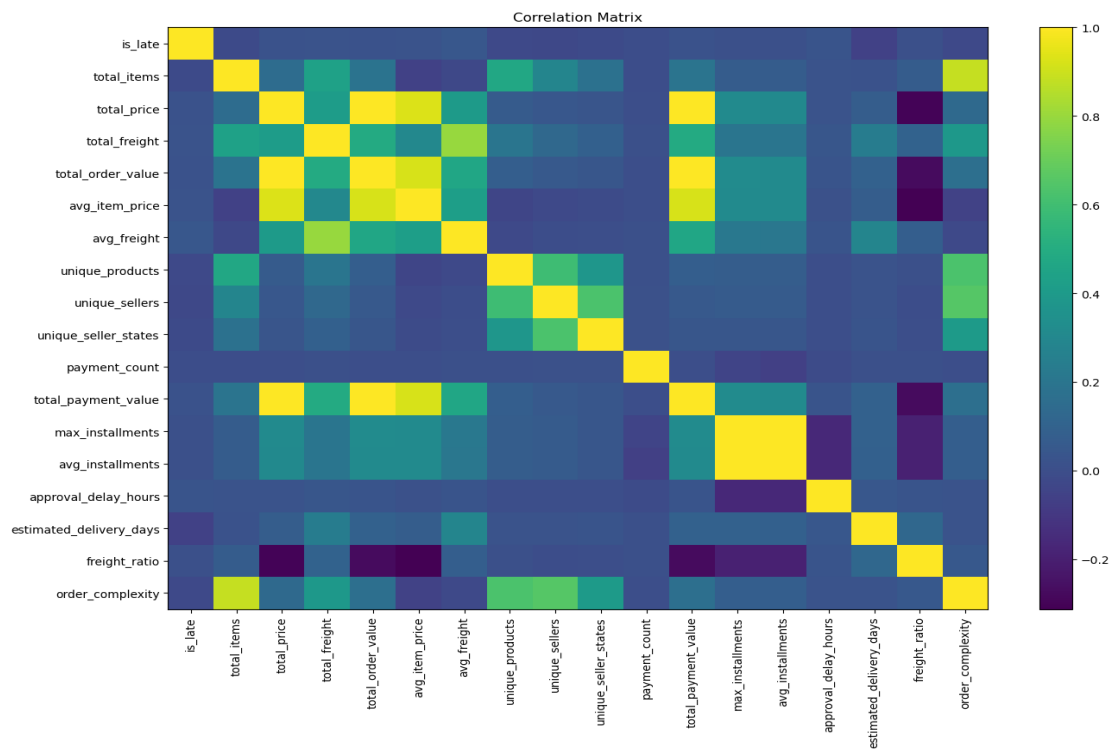
Target distribution



Late rate by month



Late rate by customer state



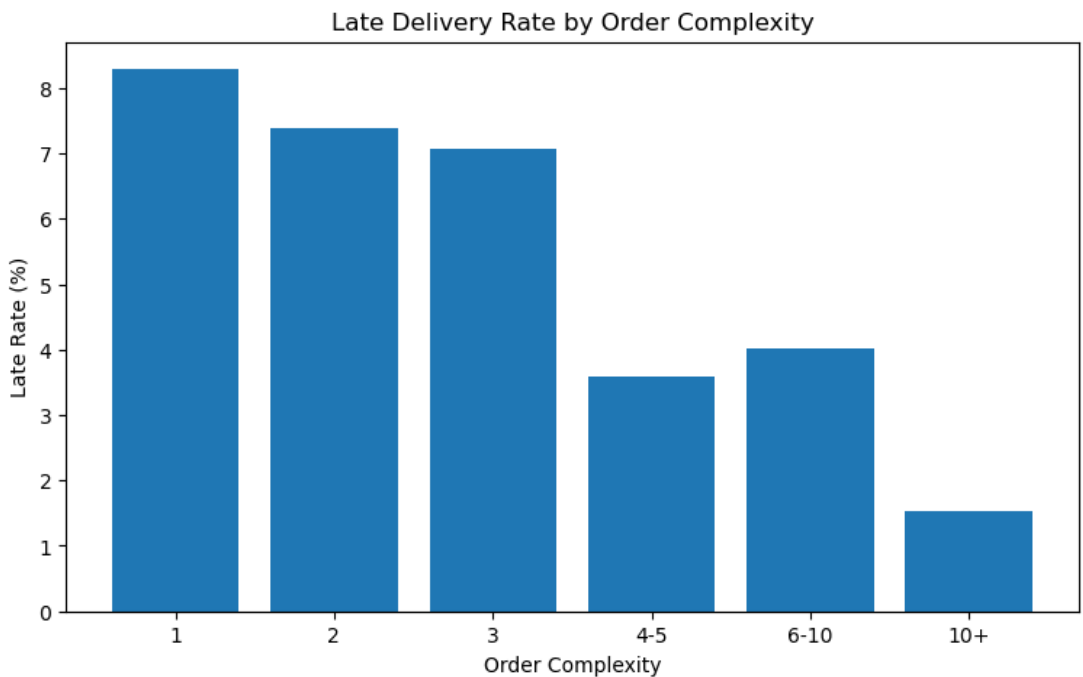
Correlation matrix

4. Time & Geographic Insights

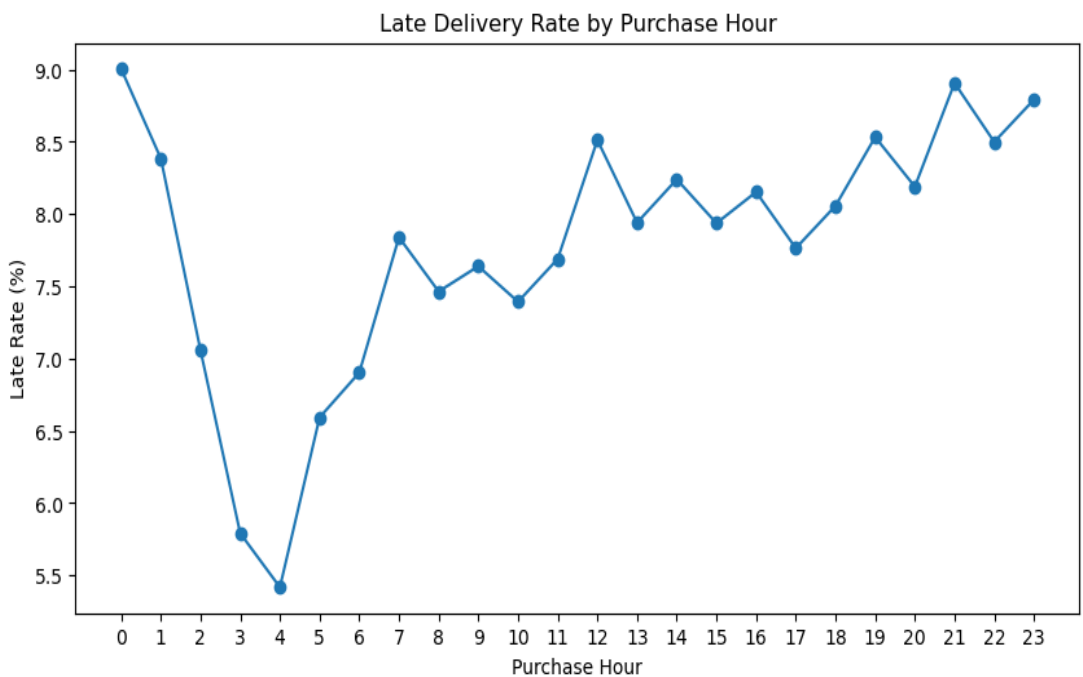
Monthly variation is substantial. March had the highest late rate at 17.15%, followed by November at 14.31% and February at 13.41%. June had the lowest rate at 2.21%. This suggests that delivery operations should not assume a constant risk level throughout the year.

Customer geography also varies strongly. Among states with at least 500 orders, MA had the highest late rate (19.67%), followed by CE (15.32%) and BA (14.04%). SP had a much lower rate (5.89%) but the largest volume, so it remains strategically important despite its lower percentage.

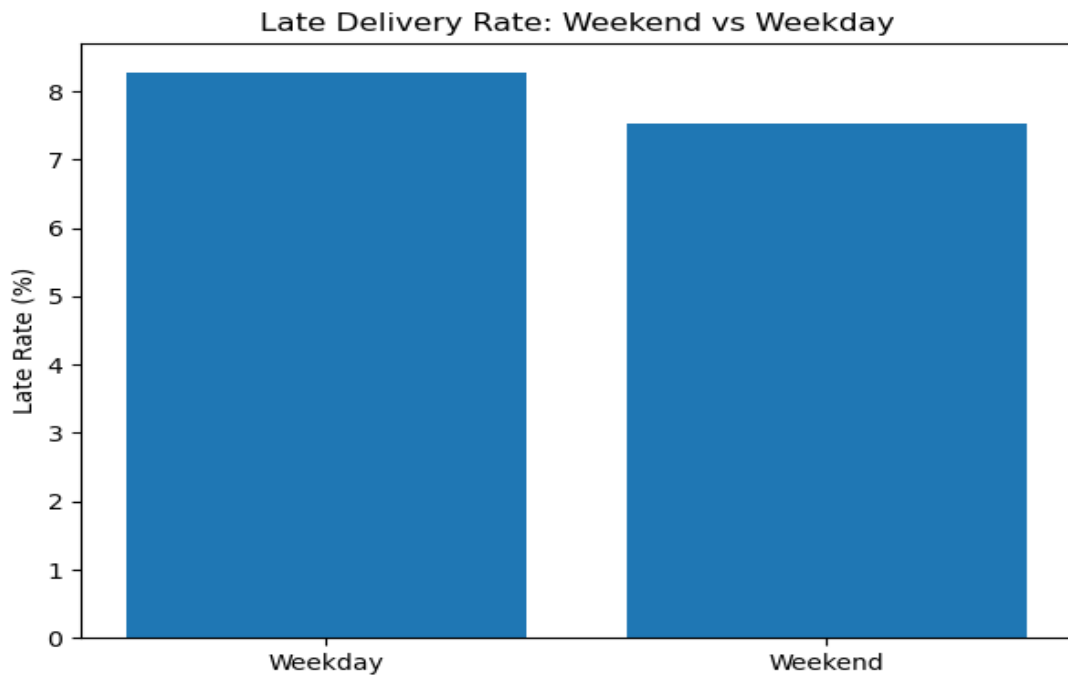
Weekend orders had a 7.54% late rate compared with 8.28% on weekdays. Purchase-hour differences were relatively modest, with rates ranging from about 5.42% to 9.00%. Therefore, month and geography appear more operationally meaningful than weekend status.



Late rate by order complexity



Late rate by purchase hour



Weekend vs weekday

5. Correlation & Feature Selection

The linear correlations with the target are weak. The strongest numerical relationship with delivery delay was `estimated_delivery_days` at -0.060, followed by `avg_freight` at 0.042 and `approval_delay_hours` at 0.029. This means simple correlation alone cannot capture the complete nonlinear and categorical structure of delivery delay. The notebook also explicitly checks for highly correlated predictors. Eight pairs have absolute correlation of at least 0.80. In particular, `total_order_value` and `total_payment_value` are perfectly correlated, while `total_price` is extremely highly correlated with both. `max_installments` and `avg_installments` are also almost identical.

Feature 1	Feature 2	Correlation
<code>total_order_value</code>	<code>total_payment_value</code>	1.000
<code>max_installments</code>	<code>avg_installments</code>	0.997
<code>total_price</code>	<code>total_order_value</code>	0.996
<code>total_price</code>	<code>total_payment_value</code>	0.996
<code>total_price</code>	<code>avg_item_price</code>	0.933
<code>total_order_value</code>	<code>avg_item_price</code>	0.921
<code>avg_item_price</code>	<code>total_payment_value</code>	0.921
<code>total_items</code>	<code>order_complexity</code>	0.885

Business/technical implication: redundant variables should be reduced when interpretability or model stability is important. For example, keeping `total_price` and dropping one of the near-duplicate `total-order/payment-value` fields can simplify the feature space without losing much information. Tree models are less sensitive to multicollinearity, while Logistic Regression benefits more directly from removing redundant predictors.

6. Delivery Delay Classification

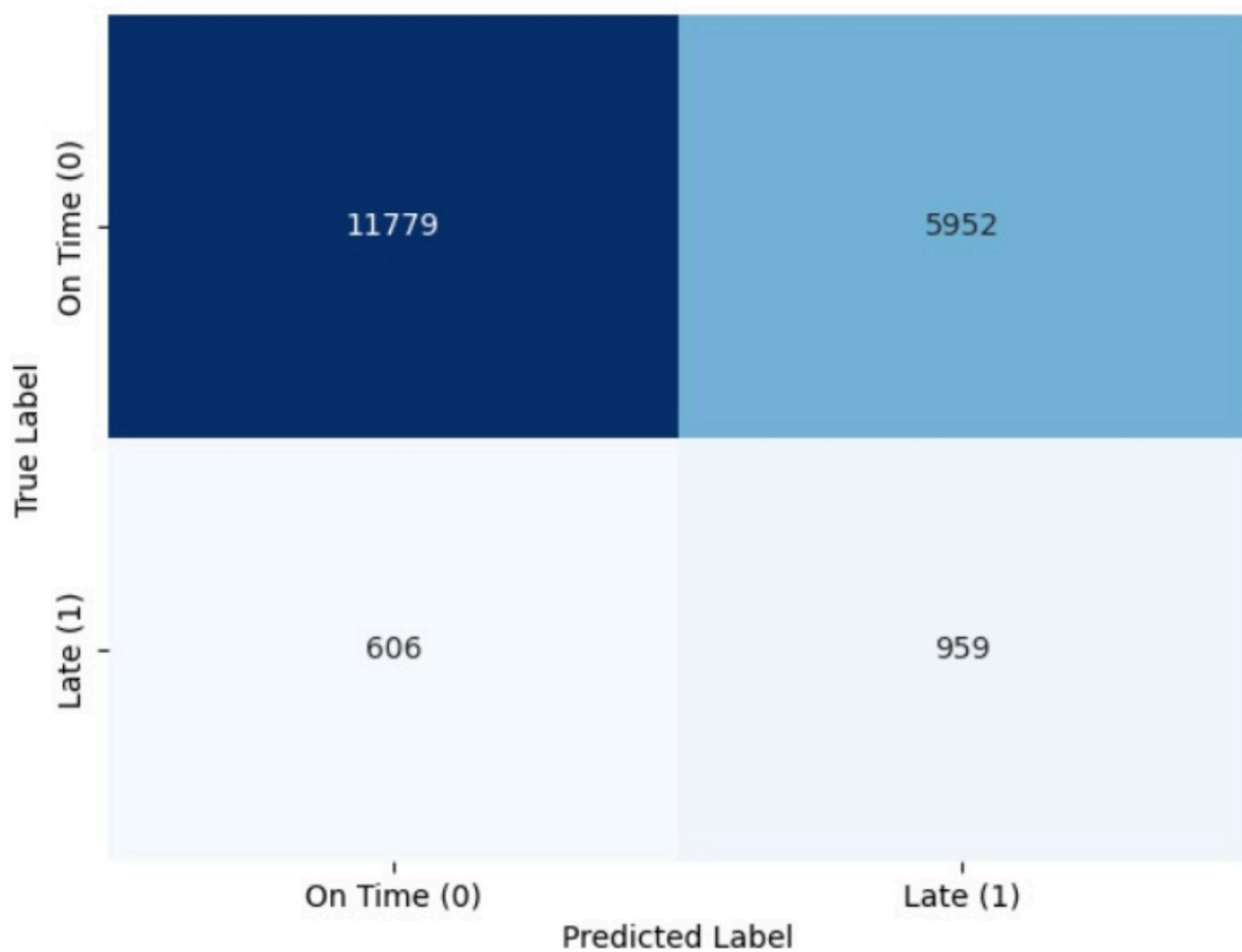
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression Balanced	66.09%	13.98%	61.73%	0.228	0.688
Decision Tree Balanced	86.53%	20.55%	23.07%	0.217	0.576
Decision Tree	87.01%	21.25%	22.24%	0.217	0.575
KNN	91.37%	29.84%	4.73%	0.082	0.606
Random Forest	91.98%	84.62%	1.41%	0.028	0.743
Random Forest Balanced	91.95%	92.31%	0.77%	0.015	0.742
Logistic Regression	91.89%	50.00%	0.38%	0.008	0.696

Because only 8.11% of orders are late, the high accuracy of the ordinary Logistic Regression and Random Forest models is misleading: their recall for late orders is extremely low. The balanced Logistic Regression is much better aligned with the business objective because it identifies 61.73% of late orders.

7. Tuned Delivery Delay Model

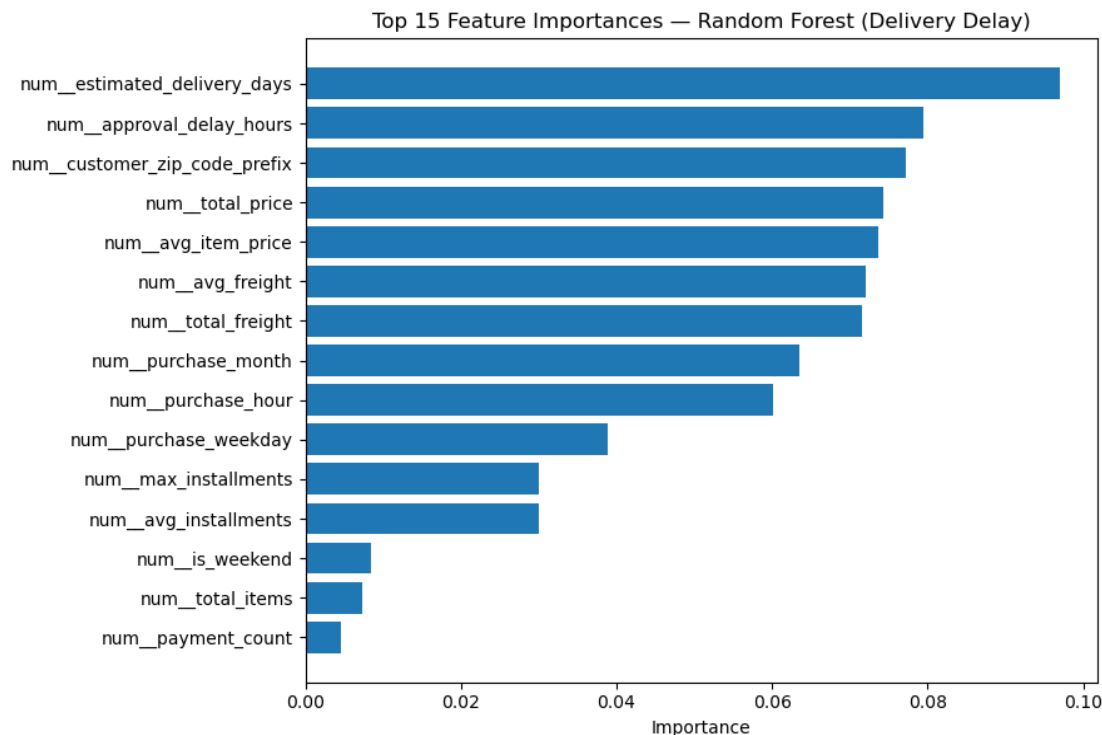
Five-fold stratified GridSearchCV tuned the balanced Logistic Regression using macro F1. The best configuration was `C=1`, L2 regularization, and the liblinear solver. Its cross-validation macro F1 was 0.5054. On the test set, precision was 0.1388, recall was 0.6128, F1 was 0.2263, and ROC-AUC was 0.6878.

Confusion Matrix - Tuned Logistic Regression



8. Feature Importance

The notebook's Random Forest feature-importance analysis identifies estimated delivery days as the strongest feature (9.69%), followed by approval delay hours (7.95%), customer ZIP-code prefix (7.73%), total price (7.42%), average item price (7.36%), average freight (7.20%), and total freight (7.16%). Purchase month and purchase hour also contribute meaningfully.



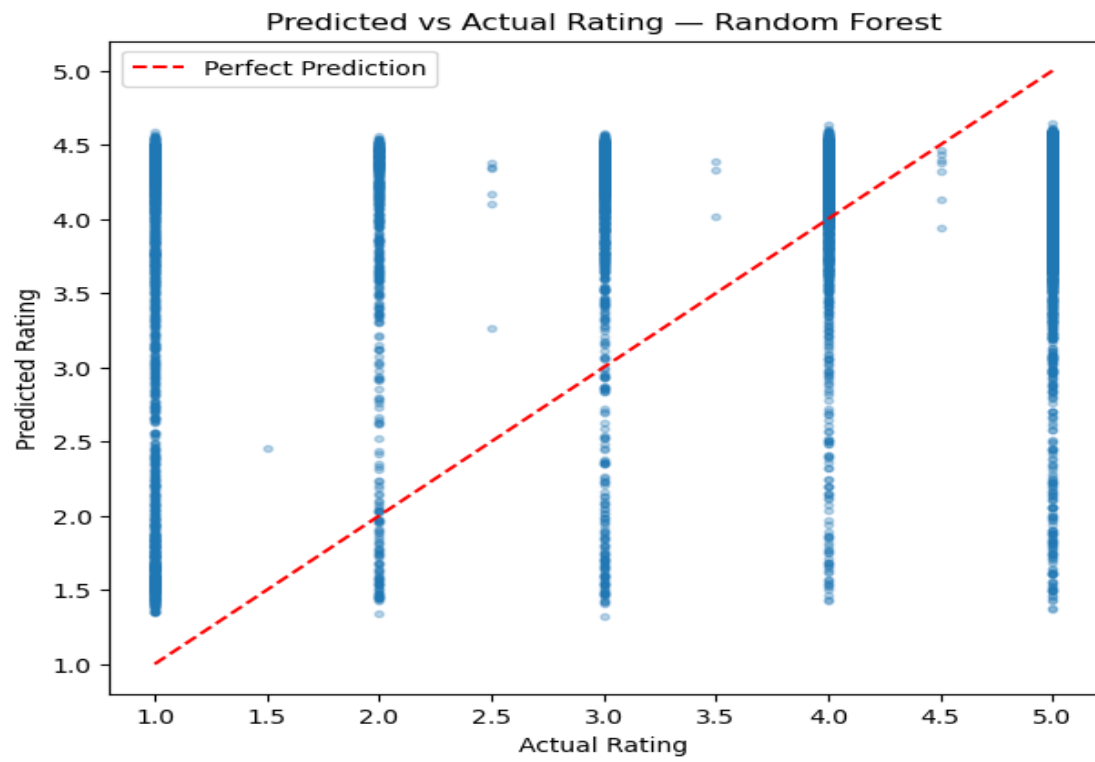
Top 15 Random Forest feature importances

9. Customer Rating Prediction

The notebook also predicts customer review ratings using 95,832 orders with review information. Four regression algorithms were compared. Random Forest Regressor performed best before tuning, with MAE 0.8640, RMSE 1.1338, and R^2 0.2149.

Model	MAE	RMSE	R ²
Random Forest Regressor	0.8640	1.1338	0.2149
Linear Regression	0.9026	1.1840	0.1439
Decision Tree Regressor	0.8862	1.1871	0.1394
KNN Regressor	0.9178	1.2529	0.0413

After tuning, the Random Forest Regressor achieved MAE 0.8662, RMSE 1.1319 and R^2 0.2175 on the test set. The improvement is modest, indicating that the available order and delivery variables explain only a limited portion of customer-rating variation.



Predicted vs actual rating and residual analysis

10. Business Insights

Problems, evidence and recommended actions for both predictive models

The following 12 insights are organized by model. Each insight connects the observed problem to evidence, business impact and a practical corrective action.

Model 1 — Delivery Delay Prediction

1. Late orders are difficult to detect because of class imbalance

Problem: Only a small share of orders are late, making the positive class harder to identify reliably.
Evidence: The tuned model achieves 61.28% recall, identifying about 61% of actual late orders.
Business impact Missed late orders can still reach customers without proactive intervention.
Fix: Use class balancing and threshold optimization to prioritize detection of late orders.

2. Short estimated delivery windows increase delay risk

Problem: Late orders receive a shorter estimated delivery window.
Evidence: Late orders average 21.97 days versus 23.89 days for on-time orders.
Business impact Short promises leave less tolerance for processing and transportation variation.
Fix: Calculate delivery promises using historical risk by state, seller, route and season.

3. Order approval delay is an important early-warning signal

Problem: Some delay risk is visible before an order enters the delivery network.
Evidence: Approval delay averages 12.31 hours for late orders versus 10.10 hours for on-time orders.
Business impact Slow approval reduces the time available to complete the promised delivery.
Fix: Introduce an approval SLA and automatically alert operations when approval is delayed.

4. Delivery risk changes significantly by season

Problem: Operational delivery risk is not constant throughout the year.
Evidence: March has a 17.15% late rate, while June has only 2.21%.
Business impact Fixed staffing and carrier capacity may be insufficient during high-risk periods.
Fix: Increase capacity, staffing and proactive monitoring before historically risky months.

5. Geographic markets have substantially different delay risk

Problem: Some states consistently show higher late-delivery rates.
Evidence: Among states with at least 500 orders, MA is 19.67%, CE 15.32%, and BA 14.04%.
Business impact Customers in these markets face materially greater delay risk.
Fix: Investigate carrier, route and seller performance in high-risk states.

6. Model alerts can be noisy

Problem: The balanced model improves recall but sacrifices precision.
Evidence: The tuned Logistic Regression identifies 61.28% of late orders, while precision is 13.88%.
Business impact Many flagged orders will not actually be late, so expensive interventions can be wasted.
Fix: Create High/Medium/Low risk tiers and reserve costly interventions for the highest-risk orders.

10. Business Insights

Model 2 — Customer Rating Prediction (continued)

Model 2 — Customer Rating Prediction

7. Customer satisfaction is only partly explained by current features

- Problem:** The model has access mainly to order and delivery variables.
- Evidence:** The tuned Random Forest achieves $R^2 = 0.2175$, so the available variables explain about 21.75% of rating variation.
- Business impact:** Improving delivery alone will not explain or solve all customer dissatisfaction.
- Fix:** Add product quality, seller quality, customer service, review text/sentiment and customer history.

8. Prediction error remains relatively large

- Problem:** Rating predictions still deviate materially from actual customer ratings.
- Evidence:** The tuned model has $MAE = 0.8662$, meaning predictions are off by roughly 0.87 rating points on average.
- Business impact:** Customer-rating predictions should support decisions rather than be treated as exact scores.
- Fix:** Add richer behavioral and service features and evaluate the model by rating segment.

9. Random Forest is the strongest tested baseline

- Problem:** The tested regression algorithms do not perform equally.
- Evidence:** Before tuning, Random Forest achieved $R^2 = 0.2149$, versus 0.1439 for Linear Regression, 0.1394 for Decision Tree and 0.0413 for KNN.
- Business impact:** Using a weaker model would provide less useful customer-satisfaction predictions.
- Fix:** Keep Random Forest as the baseline for future model development.

10. Hyperparameter tuning provides only a modest improvement

- Problem:** Changing model parameters alone does not substantially increase predictive power.
- Evidence:** R^2 improves from 0.2149 to 0.2175 after tuning, while MAE changes from 0.8640 to 0.8662.
- Business impact:** Further tuning alone is unlikely to produce a major improvement.
- Fix:** Prioritize feature engineering and new business data over extensive parameter searching.

11. Delivery performance should be analyzed together with satisfaction

- Problem:** Delivery is one contributor to customer experience, but not the only one.
- Evidence:** The relatively low R^2 shows that other factors account for most rating variation.
- Business impact:** A delivery-only strategy may leave major sources of dissatisfaction unresolved.
- Fix:** Combine delivery-risk signals with seller quality, product quality, customer service and previous ratings.

12. Customer-history and review data are key opportunities

- Problem:** The current model lacks direct information about the customer's experience.
- Evidence:** Features such as review sentiment, complaints and seller history are not represented strongly in the current prediction set.
- Business impact:** Missing experience-level information limits the model's ability to predict ratings.
- Fix:** Add review text/sentiment, seller historical rating, product quality, cancellations, complaints and service interactions.